

# AegisAI

## AI-Assisted Intelligence Analysis and Decision Support Platform

*"Transforming surveillance data into actionable intelligence."*

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### Project Overview Document

Academic Capstone Project (BSERC)

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# 1. Introduction

AegisAI is an AI-powered intelligence analysis and decision support platform, designed as an academic prototype for the BSERC project. Rather than functioning as a single-purpose object detection tool, AegisAI brings together computer vision, machine learning, deep learning, generative AI, GIS visualization, cybersecurity, and data analytics into one cohesive intelligence ecosystem.

The platform is built to support human analysts — not replace them — by processing surveillance information, organizing it into structured intelligence, identifying potential risks, generating readable reports, and presenting insights through an interactive, map-driven dashboard. The system ingests images and videos from drones, CCTV feeds, satellites, or curated datasets, and analyzes this information automatically instead of relying solely on manual observation.

The project is intended purely for research and educational purposes and is built entirely on publicly available or simulated datasets.

## 1.1 Reframing the Goal

A conventional detection project stops at a single step: upload an image, detect an object, and the task is complete. AegisAI is designed around a far richer pipeline — the system collects data, analyzes it, predicts what may happen next, explains its reasoning, recommends a course of action, compiles a report, and continues to learn from new information over time. This shift, from a single inference step to a full analysis-to-recommendation loop, is what distinguishes AegisAI from a typical CRUD application with a model attached to it.

# 2. Project Vision and Responsible Positioning

The vision for AegisAI is a research and educational platform that combines Deep Learning, Machine Learning, Generative AI, Data Analytics, GIS, and Cyber Security to transform surveillance data into actionable intelligence for human analysts.

An important design decision shapes how the platform is described and built: AegisAI is intentionally positioned as an AI-assisted Intelligence Analysis and Decision Support Platform, rather than as a system that makes military decisions on its own. Every module described in this document — including the Threat Intelligence Engine, the Predictive Intelligence module, and the Decision Support Engine — is designed to surface information, scores, and recommendations for a human analyst to review. No module is designed to autonomously issue or execute a real-world action. This framing is both technically accurate and aligned with responsible AI practice, and it makes the project a stronger fit for an academic setting.

# 3. Problem Statement

Manual analysis of surveillance images, videos, and field reports is slow and difficult to scale. Analysts reviewing large volumes of visual and textual data by hand can miss patterns, fall behind incoming data, and lack a consistent way to compare current activity against historical trends.

AegisAI addresses this by automating detection, structuring and storing intelligence in a central database, predicting emerging risks, visualizing trends spatially and over time, and generating AI-assisted narrative reports — all while keeping a human analyst in the loop for every decision.

## 4. Objectives

- 1. Detect military-related objects and assets from surveillance imagery using deep learning models.
- 2. Classify and score potential threats based on detected objects and contextual data.
- 3. Predict short-term future risk (e.g. over the next 24 hours) using historical and time-series patterns.
- 4. Provide interactive dashboards, GIS maps, and analytics for situational awareness.
- 5. Generate AI-assisted written reports and allow analysts to query the system in natural language.
- 6. Secure the application end-to-end with modern authentication, encryption, and audit logging.

## 5. Technology Stack

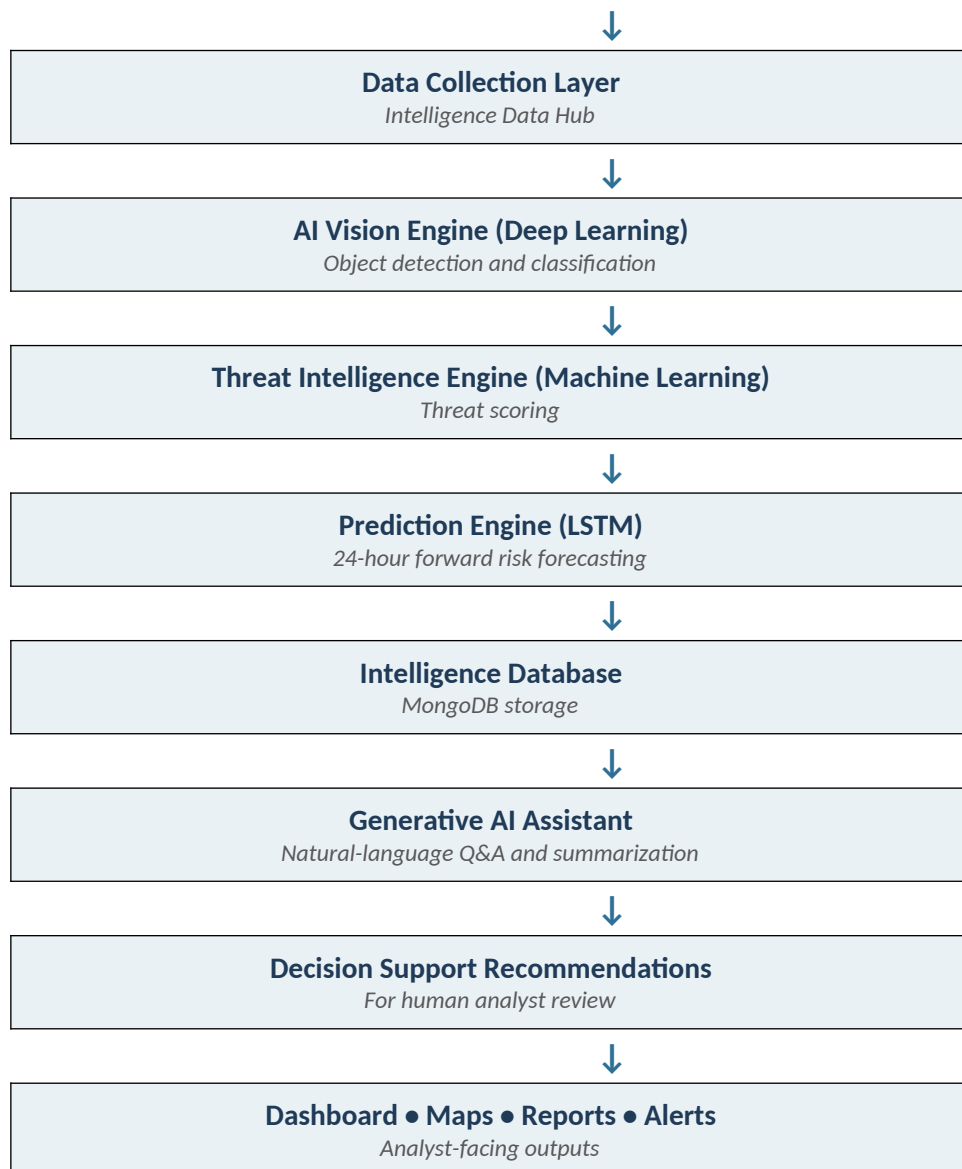
The platform integrates the following domains and technologies:

| Domain                          | Technology                             | Purpose                                          |
|---------------------------------|----------------------------------------|--------------------------------------------------|
| Deep Learning / Computer Vision | YOLOv8 / YOLOv11 / RT-DETR, OpenCV     | Detection of military-related objects in imagery |
| Machine Learning                | XGBoost, Random Forest                 | Threat scoring and classification                |
| Time-Series Forecasting         | LSTM                                   | Short-term predictive risk analytics             |
| Generative AI                   | LLM assistant (OpenAI / Llama / Gemma) | Report generation and analyst Q&A                |
| Data Analytics                  | Pandas, Plotly                         | Trend analysis and statistical insight           |
| GIS                             | Leaflet / Mapbox                       | Interactive threat mapping                       |
| Cybersecurity                   | JWT, AES, bcrypt                       | Authentication, encryption, secure APIs          |
| Backend                         | Flask / FastAPI                        | REST APIs                                        |
| Frontend                        | Next.js, Tailwind CSS                  | Analyst-facing dashboard                         |
| Database                        | MongoDB                                | Intelligence and detection storage               |

## 6. System Architecture and Workflow

Data flows through AegisAI in a linear pipeline, from raw surveillance inputs to analyst-facing outputs. Each stage enriches the data further before passing it to the next:

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| <div>Images, Videos, Reports, and Weather Data</div> <div>Drones, CCTV, satellites, patrol reports</div> |
|----------------------------------------------------------------------------------------------------------|



## 7. Core Modules

### Module 1 — Intelligence Data Hub

Collects and centralizes incoming data from multiple sources:

- Drone imagery
- CCTV imagery
- Satellite imagery
- Patrol reports
- Weather data
- Terrain data
- Historical event records

## Module 2 — AI Vision Engine

Uses a deep learning model (YOLOv8 / YOLOv11 / RT-DETR) to detect relevant assets in surveillance imagery, including soldiers, tanks, armored personnel carriers, trucks, UAVs, helicopters, and weapons. Every detection includes a class label, confidence score, timestamp, and location, and is passed on to the Threat Intelligence Engine. A representative detection output looks like this:

|             |          |
|-------------|----------|
| Object:     | Tank     |
| Confidence: | 97%      |
| Location:   | Sector A |
| Time:       | 18:35    |

## Module 3 — Threat Intelligence Engine (Machine Learning)

Combines detected objects with contextual variables — weather, time of day, historical events, region, terrain, and distance from a border — to compute a threat score and classification:

|                |               |
|----------------|---------------|
| Threat Score:  | 91%           |
| Threat Level:  | CRITICAL      |
| Risk Category: | High Priority |

## Module 4 — Predictive Intelligence

Extends the analysis from today's threat picture to a short-term forecast of what may happen next, using Random Forest, XGBoost, and LSTM time-series models:

|                            |           |
|----------------------------|-----------|
| Next 24 Hours              |           |
| Predicted UAV Activity:    | 78%       |
| Predicted Convoy Movement: | 65%       |
| Border Risk:               | Very High |

## Module 5 — Generative AI Intelligence Assistant

The natural-language heart of the platform. An analyst can ask free-form questions and receive AI-generated answers, comparisons, explanations, and full reports, for example:

- “What happened today?” → the assistant summarizes the day's detections and alerts
- “Compare this week with last week.” → the assistant generates a comparative analysis
- “Why is this threat Critical?” → the assistant explains the contributing factors
- “Generate an intelligence report.” → the assistant drafts a professional report automatically

## Module 6 — Decision Support Engine

Rather than simply stating that an object was detected, this module proposes a recommended course of action for a human analyst to evaluate and approve:

|                      |                         |
|----------------------|-------------------------|
| Threat Priority:     | HIGH                    |
| Recommended Actions: | - Increase surveillance |

- Deploy drone patrol
  - Monitor next 6 hours
  - Notify regional command

*These outputs are recommendations intended for human analysts to review and act on — the platform does not take autonomous action.*

## Module 7 — Intelligence Dashboard

Provides a real-time operational view for analysts, including:

- Live threat counter
- AI accuracy metrics
- Active alerts
- Top affected regions
- Object distribution charts
- Event timeline
- Heatmaps

## Module 8 — GIS Intelligence

An interactive map layer built on Leaflet / Mapbox that shows:

- Detected threats
- Border posts
- Patrol units
- Drone routes
- High-risk zones

## Module 9 — Cyber Security

Protects the platform and its data end-to-end:

- JWT-based authentication
- AES encryption
- Audit logging
- Role-based access control
- Secure API endpoints

## Module 10 — Automatic Intelligence Reports

Generates daily, weekly, monthly, and incident reports with a single click, exportable as PDF or DOCX.

# 8. Dataset Sources

The project relies exclusively on publicly available or simulated data:

- Drone / UAV image datasets
- Military vehicle image datasets

- Satellite imagery (e.g. Sentinel-2, Landsat)
- Weather APIs or historical weather records
- Simulated patrol reports and incidents
- Open-source object detection datasets

## 9. Project Deliverables

- Professional web application
- AI vision (object detection) module
- Machine learning threat prediction engine
- Generative AI assistant
- Interactive analytics dashboard
- GIS map interface
- Automated report generator (PDF / DOCX export)
- REST API layer
- MongoDB database
- Technical documentation
- GitHub repository
- Project presentation
- Research paper (optional)

## 10. Development Roadmap

| Phase    | Milestone                            | Status   |
|----------|--------------------------------------|----------|
| Phase 1  | Planning & Requirements              | Complete |
| Phase 2  | Architecture & UI Design             | Complete |
| Phase 3  | Frontend Development                 | Planned  |
| Phase 4  | Backend APIs                         | Planned  |
| Phase 5  | Database Integration                 | Planned  |
| Phase 6  | Deep Learning Vision Module          | Planned  |
| Phase 7  | Machine Learning Threat Prediction   | Planned  |
| Phase 8  | Generative AI Intelligence Assistant | Planned  |
| Phase 9  | GIS & Interactive Maps               | Planned  |
| Phase 10 | Cyber Security                       | Planned  |
| Phase 11 | Analytics & Report Generation        | Planned  |
| Phase 12 | Testing & Deployment                 | Planned  |

## 11. Expected Outcomes

The completed project will demonstrate how deep learning, machine learning, generative AI, data analytics, GIS, and cybersecurity can be integrated into a single intelligent decision-support platform. Rather than simply detecting objects, AegisAI converts surveillance data into structured intelligence — object detections with confidence scores, threat levels, short-term risk predictions, historical analytics, GIS visualizations, alerts, downloadable reports, AI-generated summaries, and trend analysis — that supports human analysts with visualization, reporting, and evidence-based recommendations.

## 12. Future Scope

- Real-time drone video feed integration
- Live satellite imagery integration
- Explainable AI (XAI) for greater transparency in threat scoring
- Multi-camera object tracking across sectors
- Cloud-based scaling for larger deployments
- Enhanced forecasting using additional public or simulated datasets

## 13. Closing Note

AegisAI is designed to go beyond a single-model demo. By connecting data collection, computer vision, predictive machine learning, generative AI, GIS, and security into one workflow — collect, analyze, predict, explain, recommend, report, and learn — the project aims to show, at an academic scale, how a modern intelligence-analysis system can be built responsibly, with a human analyst kept firmly in the loop at every step.